

Thermal Processes

Definition: Thermal equilibrium describes the state in which 2 or more objects have the same temperature and that there is no net transfer of energy between them.

#PhysicsDefinitions

Thermal Conduction

Definition: Conduction is a process of energy transfer where energy is transferred through the passing on of vibrational motion from one particle to another.

#PhysicsDefinitions

Conduction in solids

metals → better thermal conductors than non metals.

Although both metals and non metals are made up of atoms and molecules → metals consist of tiny particles called free electrons that are absent in most non metals.

→ electrons are extremely small compared to atoms.

due to small size and high speed → electrons can move very fast through the metal and transfer energy easily.

Process to Memorize (Non-Metals)

1. Energy is transferred to the rod by heating. particles nearest to the heat source will gain heat energy and start to vibrate more vigorously about their fixed positions.
2. Particles with more energy collide with their neighboring particles. through collisions, energy is transferred from the more energetic particles to the less energetic particles towards the right/left.
3. Hence, as more energy is transferred from left to right, particles here begin to vibrate more vigorously. temperature of this end of the rod increases.

Process to Memorize (Metals)

1. in addition to same process in non metals, free electrons at this heated end gain energy from interacting with the particles.
small size and high speed of the electrons allow them to move easily and quicker to the cooler end and transfer energy to the particles there.

2. Free electrons moving here are collectively slower after colliding and transferring their energy to the particles at this cooler end
Particles vibrate more vigorously. the temperature of this end of the rod increases.

Due the role of free electrons → materials that consist of free electrons are usually good electrical and thermal conductors.

Conduction in Liquids and Gasses

compared to solids → conduction in liquids and gasses are not as effective → as particles in liquid and gasses are spaced further apart than in solids.

→ probability of particles transferring their energy by collision is lower than in liquids and gasses.

air - poor COH

Convection

Definition: Process of energy transfer by means of convection currents of a fluid, liquid or gas, due to difference in density

#PhysicsDefinitions

Model answer for liquids (pot):

1. Liquid at the bottom of the pot expands due to heating via conduction through the bottom of the pot.
2. The warmer and less dense liquid starts sinking towards the top.
3. At the same time, surrounding dense liquid sinks to take place of the rising liquid. process repeats and causes convection currents. the circulating convection current carries warmer liquid away from the heat source and cooler liquid towards it. This allows pot to gain internal energy.

Model answer for gasses:

1. The air near the candle flame expands due to heating via conduction/radiation
2. It will gain heat and expand, thus becoming less dense, thus it will rise.
3. at the same time, the cooler air is denser, and thus it will sink to replace the air previously occupied by the warmer air.
4. the process repeats and this sets up a convection current.

→ use a cover lid to prevent a convection current from being set up.

is also affected by surface area

Radiation

both conduction and convection require a physical medium for energy transfer to occur. However radiation does not.

Definition: Radiation is the process of energy transfer by electromagnetic waves. it does not require a medium.

#PhysicsDefinitions

Radiation from sun (EM- Electromagnetic waves) → can travel through vacuum at the speed of light.

types of EM radiation → includes radio waves, visible light and others.

all objects with temperature above absolute 0 (0 kelvin or -273 degree Celsius) → emit and absorb EM radiation at the same time.

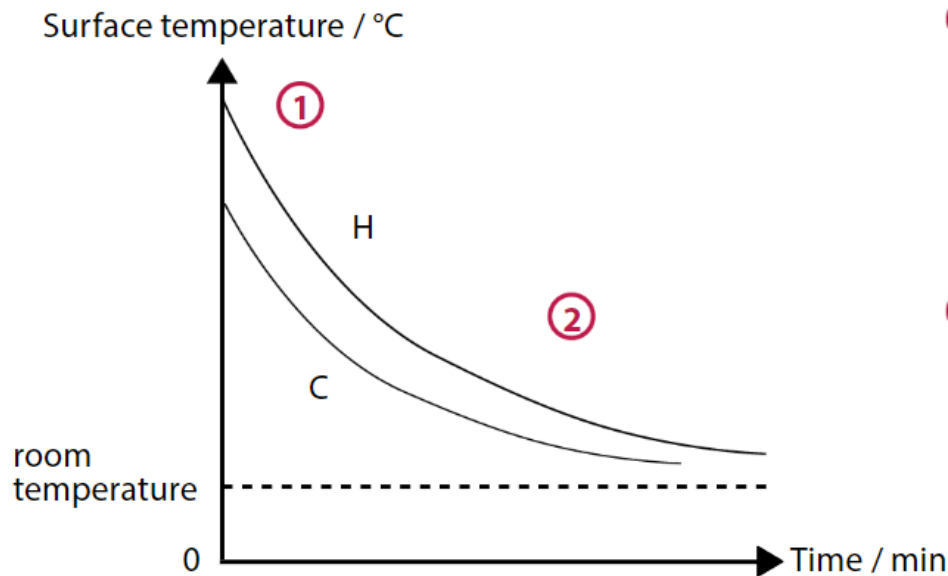
eg: plant is receiving radiation from the lamp - and it is also emitting radiation to surroundings at the same time.

Rate at which object receives radiation is affected by:

- Surface temperature
- Surface Area
- Surface Color and texture.

Surface Temperature:

- 2 identical objects that have different surface temperatures - which are both above room temperature.
- H is hotter while C is cooler.
- energy is transferred from H by radiation at a rate of R_h
- Energy is transferred from object C by radiation at a rate of R_c
- When all variables remain the same, R_h will be higher than R_c



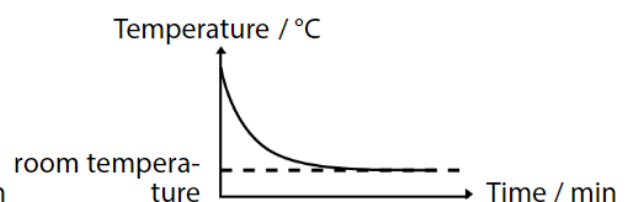
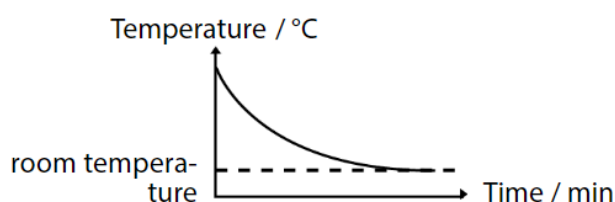
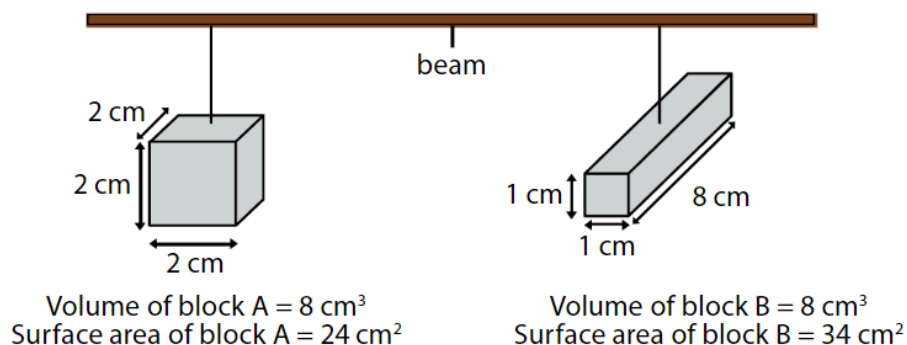
1. Initially when surface temperature is higher, the rate of temperature decrease is higher (rate of emission of radiation is higher)
2. as time passes, surface temperature decreases, rate of temperature decrease becomes lower (rate of emission of radiation is lower)

Object H → loses energy faster than object C - such that difference in their temperature becomes smaller.

- object H is still hotter and transfer energy by radiation at a higher rate than C
- higher temperature of an objects surface - higher than rate of emission of radiation.

Surface Area:

- 2 steel blocks A and B are hung from a beam
- Blocks have same initial temperature, mass and volume



Block B has a bigger surface area and cools faster than block A
Large surface area - will emit energy at a higher rate via radiation
Similarly - larger surface area will also absorb energy at a higher rate via radiation.

Surface Color and Texture:

Surface color → affects rate of energy transfer by radiation

Visible light and infrared radiation are part of the EM spectrum

Infrared radiation → is beyond red light in visible spectrum.

- we cannot see infrared radiation, but when it falls on skin, we can feel the warmth
- Red light is near infrared radiation in EM spectrum - surface that reflects red light is often a good reflector of infrared radiation.

white light source → emits light that is made up of colors in a rainbow.

Under white light source → surface will appear white when surface reflects all colors from that light source.

surface will appear yellow → if it reflects mainly yellow light - and little of other colors.
Hence white surface reflects more light than yellow surface.

Texture of surface → also affects amount of light reflected.

shiny or smooth surface - reflects more light than matt or rough surface.

Shiny, smooth, and light colored surface of an object → absorbs incoming radiation at a slower rate.

→ same surface also emits radiation at a slow rate.

poor emitter of radiation - is also a poor absorber of radiation

→ reverse is true for matt, rough, and dark colored surfaces.

- smaller surface area
- Shiny, smooth, light colored surface
⇒ slower rate of emission/absorption
- Larger surface area
- matt, rough, dark colored surface
⇒ faster rate of emission/absorption

Important note: conduction is a lot faster than radiation generally.

9.3

Applications of Thermal Processes